Global FX Intelligence - FIN623 Project Report

This report summarises the completed Global FX Intelligence project, a full-stack currency dashboard combining live exchange data, market analytics, a machine learning forecasting module, and alerting functionality. The solution was implemented as a Flask backend and a React frontend to satisfy the assignment requirements.

Project objective

The platform provides real-time FX monitoring, pair analysis, conversion tools, volatility scoring, forecasting, and local alert tracking for an academic finance dashboard case study.

Architecture

The application uses Flask Blueprints for modular API routing, SQLite for persistence, and a React UI with Tailwind for the presentation layer.

ML Forecasting

A supervised time-series regression workflow generates short-term forecasts using lag-based features and chronological data validation.

Front-end Validation

The project includes a production build check and API health validation to confirm the system is operational.

Submission Outcome

The dashboard is working, documented, and ready for live demonstration and academic review.